# Scaling of Episodic SSEs on Simulated Rate-and-State Faults: Effects of data processing and physical parameters

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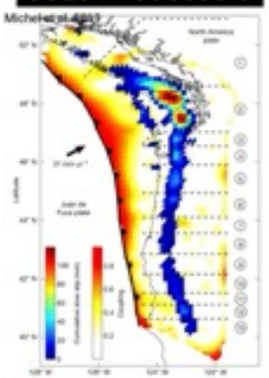
Statewide California Earthquake Center (SCEC)

## Abstract

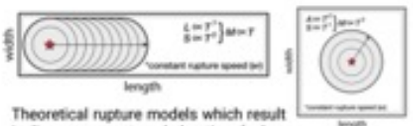
Slow-slip events (SSEs) release significant tectonic strain through aseismic slip and are observed at nearly every subduction zone across the globe. The moment-duration power law scaling for earthquakes and SSEs may reflect their source mechanisms; however, different studies find different power-law scalings. While regular earthquakes clearly follow a cubic moment-duration scaling, individual SSE catalogs observe a variable scaling ranging from linear to cubic. Studies have shown that rupture models in a rate-and-state framework are able to produce the observed cubic scaling due to variable rupture speed.

In this study, we use a Cascadia-like model to examine how the data processing assumptions and physical model parameters affect the scaling. To identify slow slip events in the simulated slip history, we need to impose a slip-rate and/or moment-rate thresholds. We find that, with a slip-rate threshold, we are able to reproduce similar moment-duration scaling with different threshold values, but increasing the threshold results in smaller, shorter events. Further limiting the area over which the moment is calculated through a moment-rate threshold changes not only the size of the events but also the scaling, ranging from sub-linear to nearly-cubic. Further, we simulate slow slip sequences for several (low) effective normal stress regimes and find that decreasing effective normal stress also decreases the power law scaling between moment and duration. Our findings indicate that while our simulations reproduce certain aspects of SSEs, the results (including the power of the moment-duration scaling) depend on how we process the simulated data as well as on the physical model parameters. These findings imply that we may be able to constrain physical properties of subduction zones where slow slip events occur, such as the effective normal stress, by matching simulations to observations, provided we carefully calibrate the processing assumptions in numerical simulations to match assumptions made in constructing the GPS catalogs. Our results further suggests that different moment-duration scaling found for SSEs in different regions and by different studies may be due to the resolution limits of the GPS networks and/or data processing assumptions.

## SSEs in Cascadia

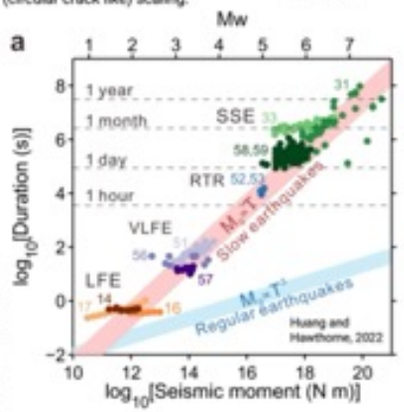


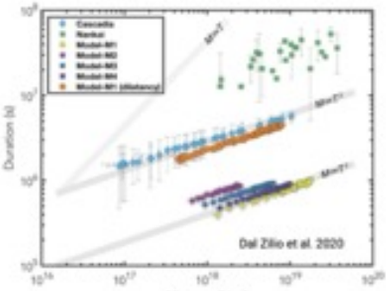
Cascadia subduction zone (CSZ) has hosted large magnitude earthquakes in the past but is currently quiescent. The presence of nearly periodic (18-22 months) SSEs give some of the only insight into the subduction dynamics. While SSE catalogs notably do not follow the high moment-low duration space of regular earthquakes, **individual catalogs (i.e. 33, 55 in right figure) show scalings more consistent with M $\alpha$ $T^3$. It's worth noting that the general trend of all SSE catalogs do, however, plot along a M $\alpha$ T, and it's unclear whether this reflects mechanical properties, source mechanisms, or instrumentation limits.**



Theoretical rupture models which result in linear moment and duration (pulse-like) versus cubic moment and duration (circular crack like) scaling.

From Dal Zilio et al. 2020




where we simulate an SSE generating fault with an elongated oval velocity weakening patch surrounded by velocity strengthening material with comparable dimensions to the Cascadia slow slip dimensions.

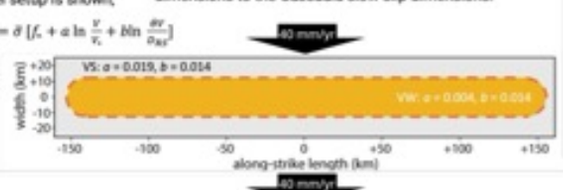
## R&S SSE Model

In this study, we use an established rate-and-state Cascadia-like quasit-static rupture simulation (following Dal Zilio et al. 2020). The rate-and-state model is able to **produce events that not only follow the cubic moment-duration scaling, but also matching the observed Cascadia catalog** distribution of events with the addition of a dilatancy parameter. A non-constant rupture speed allows for this scaling to arise. Below, our model setup is shown,

**R&S Friction:** $\tau = \bar{\sigma} f = \bar{\sigma}\left[f_* + a \ln \frac{V}{V_*} + b \ln \frac{\theta V}{D_{RS}}\right]$

**Ageing Law:** $\frac{d\theta}{dt} = 1 - \frac{V\theta}{D_{RS}}$

$f_* = 0.6$,
$V_* = 1e\text{-}6$ m/s,
$D_{RS} = 0.045$ m,
$\bar{\sigma} = 10$ MPa



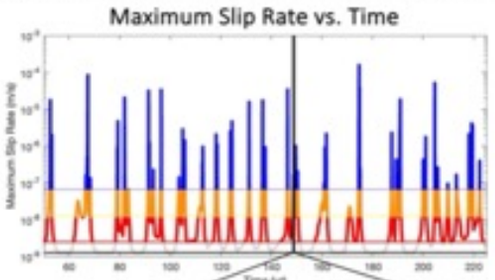

## Moment and Duration Definitions

$$M_{0,event} = \int_{t^{in}_{event}}^{t^{fin}_{event}} \dot{M}_{0,event}(t)\, dt \qquad \dot{M}_{0,event}(t) = -\sum \mu A_{s_{event}} \dot{\delta}_{event}(s_{event}, t) \text{ for } t^{in}_{event} < t < t^{fin}_{event}$$

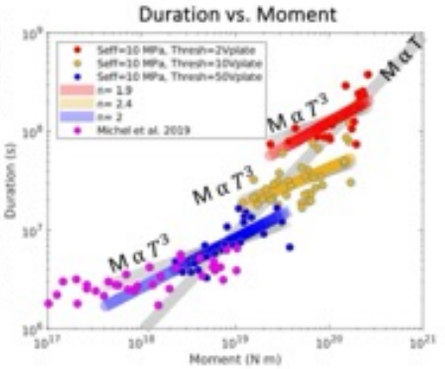
where $t^{in}_{event}$ and $t^{fin}_{event}$ represent when the slip rate, $\dot{\delta}$, on any subfault, s, surpasses and subsequently drops below $V_{thresh}$, marking the onset and end of a SSE.

## Slip Rate Threshold

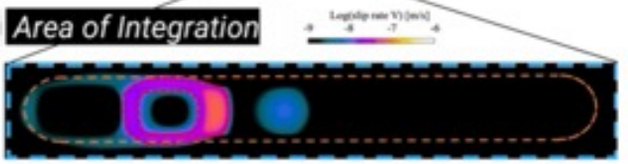
The maximum slip rate vs. time is plotted below, showing clear SSE cycles, with inter event slip rate velocities moving at plate rate in the VS region, and co-event slip rates of up to 1e-4 m/s. As is shown via color, in order to calculate an event catalog we must pick the onset and end of said event, where we apply three velocity thresholds (2x the plate rate, 10x, and 50x). Given these thresholds result in different $t_{in}$ and $t_{out}$, we see that **increasing the threshold results in smaller, events but with larger quantity**, as complexities within a given event can result in **multiple slip rate peaks that can separate into distinct events** by increasing the velocity threshold. The resulting catalogs are shown on the right, **where decreasing the velocity threshold results in more smaller magnitude and duration events, but the scaling for all three catalogs is remains best fit by a cubic power law.**

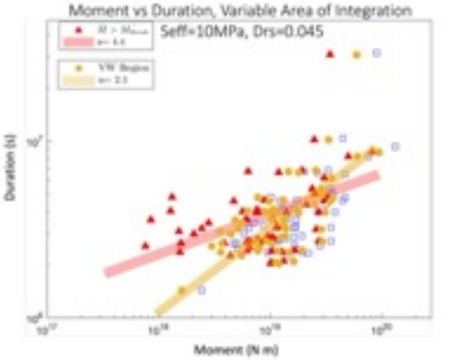



$V_{thresh1} = 2 \cdot 40\text{mm/yr}$
$V_{thresh2} = 10 \cdot 40\text{mm/yr}$
$V_{thresh3} = 50 \cdot 40\text{mm/yr}$




## Area of Integration



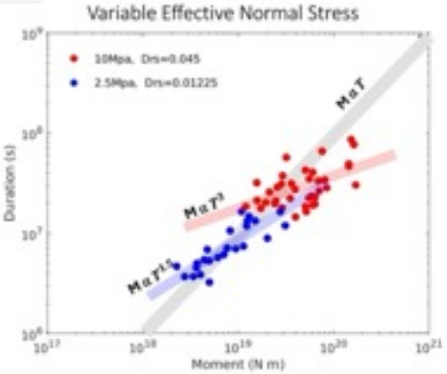

Given our simulated events fully resolve slip rate along the fault whereas geodetic determinations moment and duration are limited by the GPS array resolution, we also test the influence of our integration area on resultant scaling. With the same $t_{in}$ and $t_{out}$, integrate our slip rate first over the entire VW region and compare to the result when only considering the subfaults which surpass our the velocity threshold. The above shows an example slip rate distribution at time 148 years (black line on max slip rate plot) with orange contour corresponding to the VW region, and pink corresponding to the subfaults slipping above 10x the plate rate. For the case of effective normal stress equal to 10 MPa, **we find that while the duration remains the same (same $t_{in}$ and $t_{out}$), we find that decreasing the area particularly affects the smaller moment events such that the power law scaling changes from n=2.1 to n=4.4** (sub-cubic to greater than cubic).

## Effective Normal Stress

While we point out the dependence of our scaling on the assumptions in our data processing, we too explore the dependence of our SSE catalog scaling on the effective normal stress conditions. The effective normal stress at slow slip depths (~30km in Cascadia) are relatively unconstrained, by high Vp/Vs ratios from tomographic studies indicate high pore pressure/low Seff. Given SSEs only arise in a rate-and-state framework when the aspect ratio of the VW length to width is large (see Dal Zilio 2020 for further explanation), we vary both Seff and Drs to keep our nucleation size estimates the same (i.e. decreasing Drs by the same factor we increase Seff). We find that by decreasing effective normal stress from 10 MPa to 2.5 MPa, we decrease the smallest event size by an order of magnitude. Further, we find that the **decrease in Seff results in notably different scalings of the two catalogs, where the lower normal stress regime gives a power law scaling of n=1.5 compared to n=3 for 10 MPa.**

$$h^*_{rr} = \frac{\pi}{4}\frac{\mu^* D_{RS}}{\bar{\sigma}(b-a)}; \qquad h^*_{rs} = \frac{2}{\pi}\frac{\mu^* D_{RS}\, b}{\bar{\sigma}(b-a)^2};$$

This result indicates that, if we better understand the processing assumptions and differences between simulated and geodetic catalogs, we can not only use moment and duration scaling to understand slow slip mechanisms, but **potentially constrain physical parameters such as effective normal stress.**




## Conclusions

1. Rate and state simulations can produce properties of SSEs with observed scaling
2. However, calculated scaling is dependent on processing the data, including area, velocity etc., as well as the physical parameters assumed such as effective normal stress
3. In order to use simulations to infer physical parameters such as effective normal stress, we need to understand the impact of our processing assumptions on the resultant scaling

## References

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